

Sound Classification Using Valence and Arousal

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Abstract—This project classifies sound samples into four affective classes using arousal and valence. We used the provided preprocessed EmoSounds and IADSED datasets, mapped their target values to a common polarity scale, and compared logistic regression, decision tree, and K-nearest neighbors models. Each model was evaluated with and without SelectKBest feature selection and with and without random oversampling. The data were split 60:20:20, while the combined train and validation set was used for five-fold cross-validation and hyperparameter tuning before the untouched test set was evaluated. The cross-validation-selected EmoSounds model was logistic regression with feature selection and balancing, with test accuracy 0.675 and Macro F1 0.580. The selected IADSED model was logistic regression without feature selection or balancing, with test accuracy 0.640 and Macro F1 0.559. Overall, logistic regression was the most stable model, while several tree and distance-weighted KNN configurations showed clear overfitting.

Index Terms—sound classification, valence, arousal, affective computing, feature selection, class imbalance

I. INTRODUCTION

The main goal of this project was to classify sound samples using the polarity of arousal and valence. The problem has four classes, so each sound is assigned to one quadrant of the valence-arousal space. The provided EmoSounds and IADSED datasets also contain a large set of numeric audio features that describe dynamics, rhythm, spectral content, timbre, pitch, and tonal information. This made it possible to compare several classification methods using the same workflow.

We focused on three main questions. First, we wanted to know how well three different classifiers could separate the four affective classes. Second, we wanted to see whether reducing the input to the 20 highest-ranked features improved performance. Third, we wanted to compare the original imbalanced data with random oversampling. The report follows the project rubric by showing the mapping process, split, five-fold cross-validation, tuning setup, feature selection, balancing, train and test metrics, confusion matrices, and overfitting analysis.

The EmoSounds portion uses the same general valence-arousal affective setting described in the project materials. In this project, all reported experiments were run directly on the files provided for the course.

II. DATASET DESCRIPTION

The preprocessed EmoSounds dataset contained 600 samples and 70 columns, while IADSED contained 927 samples and 70 columns. In both datasets, two columns were arousal and valence and the remaining 68 numeric columns were used as candidate model features. There were no missing values in

TABLE I: Target Classes and Full-Dataset Counts

Class	Arousal	Valence	Emo	IADSED
1	≥ 0	≥ 0	62	164
2	≥ 0	< 0	232	482
3	< 0	< 0	101	130
4	< 0	≥ 0	205	151

either provided preprocessed file. After the mapped arousal, mapped valence, and target class columns were added, each final dataset contained 73 columns.

Table I shows the class definitions and the observed class counts. The class distributions were not balanced, especially for Class 2. EmoSounds had 232 Class 2 samples out of 600, while IADSED had 482 Class 2 samples out of 927. This imbalance was the reason we tested random oversampling instead of relying only on the original class frequencies.

III. PREPROCESSING AND EXPERIMENTAL SETUP

A. Target Mapping

The target values had to be mapped before the classes were created. EmoSounds arousal and valence were already on an approximately $[-1, 1]$ scale, so the mapping kept their original values. IADSED was centered around 5 on its original scale, so both targets were mapped using

$$x_{mapped} = \frac{x - 5}{4}. \quad (1)$$

This makes 5 equal to 0, values above 5 positive, and values below 5 negative. We then applied the four definitions in Table I. A value equal to zero was treated as nonnegative. The final files were saved as `final_emosounds.csv` and `final_iadsed.csv`.

Arousal, valence, their mapped versions, and the new target class were excluded from the model inputs so the classifier could not directly use the values that defined the label. Dominance was also excluded if present, as required by the project. The resulting feature matrix had 68 numeric audio features for each dataset.

B. Train, Validation, and Test Split

We used a stratified 60:20:20 split with random seed 42. EmoSounds was divided into 360 training, 120 validation, and 120 test samples. IADSED was divided into 555 training, 186 validation, and 186 test samples. Stratification kept the class proportions similar across the subsets.

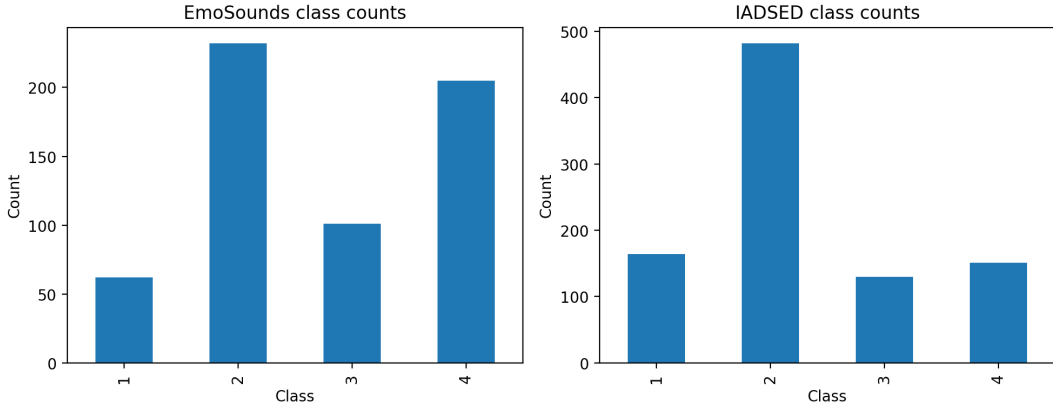


Fig. 1: Class counts before balancing for EmoSounds and IADSED. As shown in the figure, Class 2 is the largest class in both datasets.

For the rubric-required tuning step, the training and validation subsets were combined after the initial split. The combined train+validation set therefore contained 480 EmoSounds samples and 741 IADSED samples. Five-fold stratified cross-validation and grid search were performed only inside this combined development set. The test set was not used to select a model or hyperparameters.

C. Balancing and Feature Selection

The balancing method was random oversampling. It duplicates minority-class samples until each class has the same count as the largest class in the current training data. On the full train+validation portion, EmoSounds changed from class counts (50, 185, 81, 164) to (185, 185, 185, 185). IADSED changed from (131, 385, 104, 121) to (385, 385, 385, 385). Oversampling was placed inside the modeling pipeline, so in cross-validation it was learned only from the training folds and did not duplicate validation-fold examples.

For feature selection, we used univariate ANOVA F-tests through `SelectKBest(f_classif)` and retained the top 20 features. Feature selection was also inside the pipeline and therefore fit separately inside each training fold. We evaluated all four combinations for every model: no feature selection with imbalanced data, feature selection with imbalanced data, no feature selection with balancing, and feature selection with balancing.

IV. PREDICTION MODELS AND TUNING

We trained three classifiers with different levels of flexibility. Logistic regression served as a linear baseline and tuned $C \in \{0.01, 0.1, 1, 10, 100\}$. The decision tree tuned maximum depth $\in \{3, 5, 10, \text{None}\}$, minimum split size $\in \{2, 5, 10\}$, and minimum leaf size $\in \{1, 2, 4\}$. KNN tuned neighbors $k \in \{3, 5, 7, 9, 11\}$, uniform versus distance weights, and Minkowski $p \in \{1, 2\}$. Standardization was included before feature selection and modeling.

Grid search used five-fold stratified cross-validation and selected settings using Macro F1. After the best settings were found, `GridSearchCV(refit=True)` retrained the full

pipeline on the entire train+validation set. We then reported metrics on both the train+validation set and the untouched test set. This setup makes the test scores a final evaluation instead of another tuning signal.

A. Evaluation Metrics

Accuracy measures the overall fraction of correct predictions:

$$\text{Accuracy} = \frac{1}{N} \sum_{i=1}^N \mathbf{1}(y_i = \hat{y}_i). \quad (2)$$

Because the classes are imbalanced, we also used Macro F1. For class k , $F1_k = 2P_kR_k/(P_k + R_k)$, where P_k is precision and R_k is recall. Macro F1 gives each class equal weight:

$$\text{Macro F1} = \frac{1}{K} \sum_{k=1}^K F1_k, \quad K = 4. \quad (3)$$

V. RESULTS

Tables II and III report all 12 model/configuration combinations for each dataset. The reported CV value is the best five-fold Macro F1 found during tuning, and the train+validation and test columns use the refit best estimator. This gives the full comparison required by the rubric instead of reporting only the top run.

The EmoSounds model selected by cross-validation was logistic regression with feature selection and balancing. Its best C was 10, CV Macro F1 was 0.561, test accuracy was 0.675, and test Macro F1 was 0.580. The IADSED model selected by cross-validation was logistic regression without feature selection or balancing. Its best C was 1, CV Macro F1 was 0.527, test accuracy was 0.640, and test Macro F1 was 0.559.

It is important that model selection was based on cross-validation rather than the test set. For example, an EmoSounds logistic regression configuration without feature selection but with balancing happened to reach a higher test Macro F1 of 0.609. We did not call it the final best model because its CV Macro F1 was lower than the selected configuration. The same

TABLE II: EmoSounds Results for All Models and Configurations

Model	Configuration	Best hyperparameters	CV F1	Train Acc.	Train F1	Test Acc.	Test F1
Logistic	No FS / Imbal	C=1	0.533	0.823	0.766	0.700	0.566
Logistic	FS / Imbal	C=100	0.534	0.715	0.603	0.758	0.584
Logistic	No FS / Bal	C=100	0.537	0.873	0.854	0.692	0.609
Logistic	FS / Bal	C=10	0.561	0.654	0.616	0.675	0.580
Tree	No FS / Imbal	depth=5, split=2, leaf=4	0.507	0.798	0.702	0.592	0.410
Tree	FS / Imbal	depth=10, split=2, leaf=1	0.475	0.990	0.988	0.600	0.481
Tree	No FS / Bal	depth=None, split=2, leaf=4	0.518	0.898	0.888	0.575	0.464
Tree	FS / Bal	depth=5, split=2, leaf=2	0.520	0.727	0.695	0.558	0.452
KNN	No FS / Imbal	k=11, w=dist, p=2	0.527	1.000	1.000	0.675	0.569
KNN	FS / Imbal	k=5, w=unif, p=1	0.539	0.727	0.656	0.675	0.546
KNN	No FS / Bal	k=9, w=dist, p=1	0.468	1.000	1.000	0.550	0.531
KNN	FS / Bal	k=9, w=dist, p=2	0.521	1.000	1.000	0.550	0.524

TABLE III: IADSED Results for All Models and Configurations

Model	Configuration	Best hyperparameters	CV F1	Train Acc.	Train F1	Test Acc.	Test F1
Logistic	No FS / Imbal	C=1	0.527	0.726	0.667	0.640	0.559
Logistic	FS / Imbal	C=1	0.491	0.650	0.539	0.608	0.514
Logistic	No FS / Bal	C=0.01	0.496	0.590	0.581	0.597	0.576
Logistic	FS / Bal	C=1	0.476	0.521	0.506	0.554	0.547
Tree	No FS / Imbal	depth=10, split=5, leaf=1	0.440	0.874	0.851	0.559	0.462
Tree	FS / Imbal	depth=5, split=10, leaf=1	0.438	0.711	0.605	0.597	0.461
Tree	No FS / Bal	depth=5, split=10, leaf=1	0.406	0.594	0.608	0.430	0.407
Tree	FS / Bal	depth=5, split=10, leaf=1	0.444	0.507	0.532	0.430	0.448
KNN	No FS / Imbal	k=11, w=dist, p=2	0.462	1.000	1.000	0.651	0.547
KNN	FS / Imbal	k=3, w=dist, p=2	0.496	1.000	1.000	0.565	0.505
KNN	No FS / Bal	k=3, w=dist, p=2	0.427	1.000	1.000	0.554	0.523
KNN	FS / Bal	k=3, w=dist, p=2	0.464	1.000	1.000	0.495	0.486

issue occurred in IADSED, where a balanced logistic model reached test Macro F1 0.576 but had weaker CV performance. Keeping the test set separate avoids choosing a model after seeing its final test result.

A. Feature Selection and Balancing Comparisons

Table IV summarizes the average performance across the three model families for each of the four configurations. For EmoSounds, feature selection produced a small average improvement in test Macro F1 when the results are averaged across balancing choices: 0.528 with feature selection compared with 0.525 without it. For IADSED, feature selection reduced average test Macro F1 from 0.512 to 0.493. The effect was therefore dataset-dependent rather than consistently positive.

Balancing also had different effects. For EmoSounds, random oversampling barely changed average Macro F1, from 0.526 without balancing to 0.527 with balancing, while average accuracy decreased from 0.667 to 0.600. For IADSED, balancing reduced average accuracy from 0.603 to 0.510 and average Macro F1 from 0.508 to 0.498. This shows that making the training classes equal did not automatically improve generalization.

For the selected EmoSounds pipeline, the 20 ANOVA-ranked features included RMS dynamics measures, spectral centroid variation, several MFCC standard deviations, spectral novelty, rolloff, spread, low-energy variation, and spectral flux measures. The complete selected feature list is available directly from the executed notebook, which prints the feature names after the final model is chosen. The selected IADSED

model did not use feature selection, so all 68 audio features were retained.

B. Confusion Matrices

Figure 2 shows the test confusion matrices for the two CV-selected models. In EmoSounds, the model correctly classified 37 of 47 Class 2 test samples, 14 of 20 Class 3 samples, and 28 of 41 Class 4 samples. Class 1 was the most difficult: only 2 of 12 Class 1 samples were correct, and 9 were predicted as Class 2. There was also noticeable confusion between Classes 3 and 4.

For IADSED, the strongest diagonal count was again Class 2, with 77 of 97 samples correct. Class 1 had 13 of 33 correct, Class 3 had 12 of 26, and Class 4 had 17 of 30. The confusion matrix therefore shows why accuracy alone is not enough. Both selected models performed much better on the largest class than on at least one minority class, which is reflected in the lower Macro F1 values.

C. Overfitting Analysis

The train and test results show that model complexity mattered. Several distance-weighted KNN runs reached 1.000 train+validation accuracy and Macro F1 because a training point can be its own nearest neighbor, while their test Macro F1 values were much lower. This is a clear sign that the perfect development-set predictions did not generalize. Some decision tree configurations also had very large gaps, including EmoSounds feature-selected imbalanced trees with train Macro F1 0.988 and test Macro F1 0.481.

Logistic regression showed smaller gaps and the strongest CV scores on both datasets. This is one reason the CV selection process chose logistic regression for both final confusion

TABLE IV: Average Results Across Model Families for the Four Required Configurations

Dataset	Feature Sel.	Data	Train Acc.	Train F1	Test Acc.	Test F1
EmoSounds	No FS	Imbal	0.874	0.823	0.656	0.515
EmoSounds	No FS	Bal	0.924	0.914	0.606	0.535
EmoSounds	FS	Imbal	0.810	0.749	0.678	0.537
EmoSounds	FS	Bal	0.794	0.770	0.594	0.519
IADSED	No FS	Imbal	0.867	0.839	0.616	0.523
IADSED	No FS	Bal	0.728	0.730	0.527	0.502
IADSED	FS	Imbal	0.787	0.715	0.590	0.493
IADSED	FS	Bal	0.676	0.679	0.493	0.494

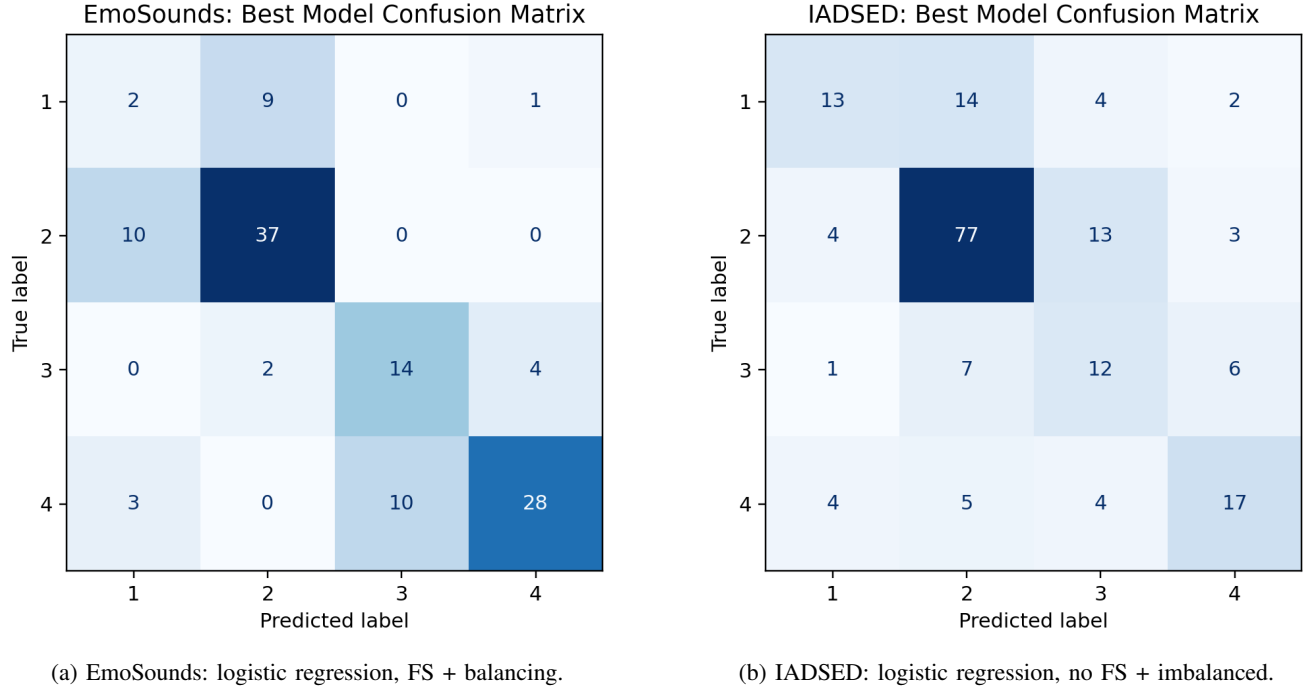


Fig. 2: Test-set confusion matrices for the models selected by five-fold CV Macro F1.

matrices. The results suggest that the audio features contain useful class information, but the sample sizes and overlap among the affective classes make highly flexible models easy to overfit.

VI. CONCLUSION AND FUTURE WORK

This project classified sound samples into four valence-arousal classes using EmoSounds and IADSED. We mapped the targets, created and saved the required final datasets, used a stratified 60:20:20 split, ran five-fold cross-validation on train+validation, tuned three classifiers, applied ANOVA feature selection, tested random oversampling, and evaluated every required configuration with accuracy and Macro F1.

The main result is that logistic regression was the most reliable model in this experiment. The selected EmoSounds pipeline reached 0.675 test accuracy and 0.580 Macro F1, while the selected IADSED pipeline reached 0.640 accuracy and 0.559 Macro F1. Feature selection helped EmoSounds slightly on average but reduced IADSED performance. Random oversampling did not produce a consistent improvement and was especially weak on IADSED. The confusion matrices

also showed that the largest class was easier to recognize than some minority classes.

The results could be improved in several ways. A future experiment could tune the number of selected features instead of fixing it at 20, try class-weighted losses as an alternative to duplicating samples, or evaluate additional models such as support vector machines or ensemble methods. Nested cross-validation would also provide a stronger estimate when the dataset is small. The project sheet lists the benchmark accuracy and F1 values as “TBD,” so there was no numeric benchmark available to verify against in the provided rubric materials.

SUBMISSION NOTE

Overleaf project link:

<https://www.overleaf.com/read/nptpgpycwtzw#4a148c>